

Release History

Notable changes between versions

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ABSTRACT

This document records the user-visible changes to the project. The format is loosely based on the Keep a Changelog convention; versioning follows Semantic Versioning.

1. Unreleased

- Semantic search using TF-IDF vectors as lightweight embeddings.
- Search analytics tracking with query history and statistics.
- Query autocomplete API with real-time suggestions.
- Document management API (add/remove documents via API).
- Related queries feature to discover similar search terms.
- Enhanced web UI with autocomplete and related queries.
- New API endpoints: `/api/semantic-search`, `/api/autocomplete`, `/api/analytics`, `/api/top-queries`, `/api/related-queries`.
- Updated version to 0.3.0 with new features.

2. Version 0.2.0 — May 2026

Added

- Brand identity: SVG logo and favicon.
- Modular CSS bundle (base, components, home, results).
- `/about` page describing the project and API.
- `/healthz` liveness endpoint.
- Five black-and-white SVG architecture diagrams.
- Five reference PDFs replacing the previous markdown docs.

Changed

- Polished search form, results page, and result cards.
- Sidebar on the results page now shows the parsed query, index stats, and operator help.
- Redesigned hero with a feature grid and a how-it-works strip.

3. Version 0.1.0 — May 2026

Added

- Core NLP preprocessing.
- Inverted index with positions and per-document statistics.
- TF-IDF (cosine) and Okapi BM25 ranking models.
- Query parser supporting `+must`, `-must-not`, `"phrase"`, and boolean operators.
- Snippet generator with highlighting.
- Damerau-Levenshtein spell suggestions.
- IR metric helpers: `precision/recall@k`, `MAP`, `MRR`, `nDCG@k`.

- CLI with one-shot and REPL modes.
- Flask web UI with HTML pages and a JSON API.
- Pytest suite with 25 tests.
- Sample corpus of seven NLP-themed documents.